GG-T-258a
September 27, 1967
SUPERSEDING
Int. Fed. Spec. GG-T-00258 (GSA-FSS)
July 8, 1964 and
Fed. Spec. GG-H-941c
October 15, 1958

FEDERAL SPECIFICATION

TESTER, BATTERY ELECTROLYTE SOLUTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers battery electrolyte solution testers for use in determining the specific gravity of storage battery electrolyte at temperatures and ranges specified.

1.2 Classification. Testers shall be furnished in the following classes and styles, as specified (see 6.2).

Class 1. Testers with the thermometer an integral part.
- Style A. Specific-gravity range 1.130 to 1.310-automotive and other.
- Style B. Specific-gravity range 1.100 to 1.350-automotive and other.
- Style C. Specific-gravity range 1.160 to 1.250-aircraft.
- Style D. Specific-gravity range 1.210 to 1.300-aircraft.

Class 2. Testers with the thermometer separate.
- Style A. Specific-gravity range 1.130 to 1.310-automotive and other.
- Style B. Specific-gravity range 1.100 to 1.300-portable and radio.
- Style C. Specific-gravity range 1.120 to 1.300-submarine.
- Style D. Specific-gravity range 1.060 to 1.240-submarine.
- Style E. Specific-gravity range 1.200 to 1.280-submarine.

Class 3. Testers without thermometer.
- Style A. Specific-gravity range 1.130 to 1.310-automotive and other.
- Style B. Specific-gravity range 1.150 to 1.400-measuring mixing acid.
- Style C. Specific-gravity range 1.200 to 1.400-automotive and other.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

- PPP-B-566 - Boxes, Folding, Paperboard.
- PPP-B-585 - Boxes, Wood, Wirebound.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Box, Fiberboard.
- PPP-C-843 - Cushioning Material, Cellulosic.

Federal Standard.

- Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified, the contractor shall furnish a complete tester sample for preproduction inspection in accordance with 4.1 (see 6.2). The preproduction sample shall be identical with the proposed production items, and shall be made and assembled by the same methods that are to be used in production of the items to fulfill the terms of the contract. Unless otherwise specified, one sample shall be furnished (see 6.2). The approved sample and the preproduction report shall be used as the production standard.

3.2 Design. The tester shall be new. The tester set shall include a float and syringe. For classes 1 and 2, the tester set shall include a thermometer and correction scale. All testers covered by this specification shall be designed for use in measuring specific gravities of electrolyte in storage batteries of the automotive, submarine, aeronautical, and other types, except batteries in which the variation of specific gravity from the charged to discharged condition is less than 50 points (0.050).

3.2.1 Construction. Construction of the instrument shall be rugged, to allow for rough handling both in shipment and in service and shall be free from any characteristics or defects that may render the instrument incapable of performance of any of the requirements specified herein.

3.2.2 Materials. The materials shall be of the quality necessary to produce testers (suitable for the intended purpose) and shall be in accordance with the requirements specified herein.

3.2.3 Interchangeability. All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced or adjusted. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number. For deliveries to military agencies, the marking shall be in accordance with MIL-STD-130.

3.2.4 Definition, tester. The term "tester" as used throughout this specification shall mean the complete assembly (see 3.3). The term "float" as used throughout this specification shall mean the bulb and calibrated stem usually referred to as a tester (see 3.3).

3.3 Component parts. The tester shall consist of an assembly of the following parts (see 3.11.1).

(a) A float of the fixed-weight variable-volume type (see 3.3.1).
(b) A thermometer to be used in determining temperature corrections (provided with class 1 and class 2 instruments) (see 3.6).
(c) A syringe consisting of a glass cylinder (in which the float is housed), depending upon the class and style required, a rubber bulb closing the upper end of the cylinder, a rubber plug or stopper to prevent the float from entering the bulb. a rubber base or cup with inlet tube closing the lower end, and such other parts as may be necessary to protect the instrument from breakage and prevent excessive dripping, or otherwise to facilitate the use of the instrument (see 3.4, 3.5.1, and 3.5.2).
(d) A tester case (see 5.1.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

3.3.1 Float. The float shall be made of high-grade annealed transparent glass, and shall be free from blisters, spots, and defects which may affect the operation and life of the instrument. It shall consist of an airtight glass tube float having a cylindrical bulb in the lower portion of which a ballast shall be securely fixed in position. The float shall have a cylindrical extension called the stem which is coaxial with the bulb. There shall be no deviation from axial alignment of the bulb and stem apparent to the eye. The stem shall contain a graduated scale and maintain a vertical position when floating freely in an electrolyte of the highest specific gravity for which it is intended. The stem shall not deviate by more than 1/16 inch per inch of the length of the scale, when the float is tested as specified in 4.5.3.2. Replacement data shall be marked on each float scale (see 3.3.2), showing the manufacturer's name and number of the float, temperature range, specific gravity, and any other information required to facilitate replacement.

3.3.2 Float scale. The float scale shall be graduated to read specific gravity at 80° F. (see 3.3.3). The scale shall be neatly printed in black on high grade white paper, high grade buff paper, or plastic. The scale divisions shall be numbered uniformly in groups convenient for easy reading, as 1.150, 1.200, 1.250, etc., or as 1.160, 1.200, 1.240, etc. The numbers shall appear in not less than two places (sides) on the scale, except the class 2, styles C and D on which the numbers shall appear in not less than three places (sides) on the scale. The graduated length of the scale shall be not less than one third (1/3) the overall length of the float (see table I). Each scale division shall represent 5 points (0.005) specific gravity, with the following exceptions: for the class 1, style B, and class 3, style C tester floats, each scale division shall represent 10 points (0.010) specific gravity.

TABLE I. Float characteristics

	Small float	Large float
Overall length of float, inches	3-1/2 to less than 4-1/2	4-1/4 to 6-3/4
Maximum number of divisions per inch of graduated scale	20	22
Milliliters of electrolyte required to float tester float at the 1.000 to the 1.210 mark, not more than	25	50
At the 1.250 to the 1.400 mark, not more than	10	35

3.3.3 Float accuracy. Floats shall be accurate to within plus or minus 5 points (0.005) specific gravity at 80° F. with the following exceptions: the class 1, styles C and D and the class 2, styles C, D, and E floats shall be accurate to within plus or minus 2.5 points (0.0025) specific gravity at 80° F. (see 4.5.3.4).

3.4 Cylinder. When tested as specified in 4.5.10, the cylinder or barrel of the instrument which houses the float shall be of clear glass which shows no signs of ring strain and no sharp longitudinal strain when viewed under a polariscope, and shall be free of defects which will affect operation and life of the instrument. The length of the float chamber shall provide for free movement of the float in an axial direction not less than 1 inch. The inside diameter of the cylinder shall allow free motion of the float without requiring a greater volume of liquid than that specified in table I. Provision shall be made for the float to assume a free floating position within the limits of its scale without adhering to the walls of the cylinder. Cylinders that house and hold the thermometer and correction scale shall hold them in a manner to protect them from breakage and deterioration caused by moisture and acid (see 3.5.2). Cleaning of barrel, also its assembly with other parts of the instrument, shall be easily accomplished. There shall be no evidence of cracking or shattering of any of the elements of the tester, when tested as specified in 4.5.7.

3.4.1 Glass cylinder. For all instruments, the cylinder or barrel of the instrument which houses the float shall be of clear glass. The glass cylinder shall have wall thickness in inches as specified in table II. The ends of the cylinder shall be rounded, reinforced and fire polished. Construction of class 1, style B testers shall be such that when the instrument is completely assembled the float shall not contact the glass barrel at any point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

TABLE II. Wall thickness

Outside diameter of cylinder	Wall thickness	Wall thickness, tolerance
Inches	Inch	Inch
5/8 ± 1/32	0.040	± 0.006
3/4 ± 1/32	.044	± .006
7/8 ± 1/32	.048	± .006
1 ± 1/32	.052	± .006
1-3/8 ± 1/32	.0625	± .006

3.5 Rubber fittings, for each (tester) assembly.

3.5.1 Upper fittings. Upper fittings shall consist of a bulb and a plug or stopper. The bulb capacity shall be such as to insure sufficient electrolyte being drawn into the barrel to obtain an accurate specific-gravity reading at any point on the float scale. The rubber in the bulb shall be of such formulation that the bulb will be pliable at all temperatures within the thermometer scale range of the instrument. An eyelet shall be molded in the bulb for hanging up the instrument. A plug of suitable material shall be in the bulb end of the barrel to prevent the float from entering into the bulb. Construction of class 1, style B testers shall incorporate an overflow tube to prevent drawing electrolyte into the float chamber to a point where the top of the float contacts the plug in the top of the barrel.

3.5.2 Lower fittings. Lower fittings shall consist of a base or cup to close the lower end of the cylinder and an inlet tube. The base shall be made of suitable material and shall be securely fitted to the lower end of the barrel. The inlet tube shall be either an integral part of, or securely fastened to, the base. All inlet tubes shall be made of soft (pliant) rubber, except the class 2, style B inlet tube which shall be made of soft or hard rubber, as specified (see 6.2). Unless otherwise specified, the inlet tube shall be of the dimensions specified in table III (see 3.11 and 6.2). The lower fittings shall be so designed as to permit the free expulsion of liquid from the cylinder irrespective of the position of the float. In no case shall the liquid expended differ in amount from the previous intake, when all parts are initially dry, by more than 1 ml. when tested as specified in 4.5.3.3. Nondrip features are desirable, provided the total liquid retained does not exceed the limit specified above. Lower fittings may contain the thermometer and correction scale as an option to 3.4 requirements.

3.5.3 Function of fittings. The upper and lower fittings shall remain operative, function properly and prevent leakage in use at any temperature within the range of the instrument. Either the bulb or the base or both shall be designed to prevent the instrument from rolling when placed on a smooth surface. Fittings shall serve as a protection to the glass parts. When the completely assembled instrument is laid on a flat surface, no glass parts shall touch that surface. Fittings shall not roughen nor shall there be any sediment in the test solution when tested as specified in 4.5.5.

3.6 Thermometers. Thermometers for classes 1 and 2 instruments shall be used to indicate the corrections to be applied to specific-gravity readings when measurements are made at temperatures other than 80° F.

3.6.1 Correction basis. The corrections shall be based on the assumption that the specific-gravity changes by one point (0.001) for each 2-1/2° F. The corrections are to be added when the temperature of electrolyte is above 80° F. and to be subtracted when below 80° F.

3.6.2 Correction scale, class 1 thermometer. Thermometers for class 1 testers shall have a specific-gravity correction scale in divisions, each representing not more than two points of specific gravity (0.002); and numbered in groups of not more than two divisions above and below the division corresponding to a temperature of 80° F. The correction scale shall be accurate to one division (0.002 specific-gravity). Specific-gravity correction figures that are to be added shall be black, those to be subtracted shall be red; markings shall be easily readable, and numbers shall be horizontally bisected by the lines coinciding with their values or numbers shall be immediately above the lines coinciding with their values. The scale and chart shall be made of plastic or good quality paper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

3.6.3 Correction scale, class 2 thermometers. Thermometers which are supplied as separate instruments, class 2, shall be graduated with correction scales as provided in 3.6.2 except that the correction scale shall be graduated in divisions representing one point (0.001), accurate to one division and numbered in groups of not more than five divisions.

3.6.4 Temperature range, (scale) Fahrenheit. The thermometers shall include the range, 20° to 120° F. with corresponding corrections, with the following exceptions: class 1, style B instruments, the thermometer shall include the range minus 65° to plus 165° F., class 2, styles B, C, and D, which shall include the temperature range 20° to 150° F.

3.6.5 Thermometer accuracy, class 1 and class 2. The thermometers of the class 1 and class 2 testers shall be accurate to within ± 5° F. in comparison with standardized thermometers, when tested as specified in 4.5.4. The temperature differential between first draw and equilibrium shall be not greater than 5° F. for the thermometer of the class 1 tester when tested as specified in 4.5.9. The thermometer of the class 2 tester which is furnished separately shall indicate equilibrium values, provided that the bulb of the thermometer is completely immersed in the solution to be measured and that it remains there for a sufficient time for equilibrium to be established.

3.7 Chart for class 1, style B. A chart shall be furnished (see 5.1.1) showing freezing points of fully and partially charged batteries.

3.8 Temperature and humidity conditions.

3.8.1 Storage temperature (for military agencies). The class 1, style B tester with the case and tester enclosed, and all components therof, shall conform to the accuracy requirements of 3.3.3 and 3.6.5 and shall show no evidence of damage when tested as specified in 4.5.1.1 and 4.5.1.2.

3.8.2 Operating temperatures. The tester and all components therof, shall be able to withstand without damage subjection to any temperature within the range of its temperature scale. Except for the class 1, style B tester, each tester shall operate within the accuracy and performance requirements set forth in 3.3.3, 3.5.1, and 3.6.5, while being subjected to any temperature within the range of its temperature scale. As a minimum requirement the class 1, style B tester shall fulfill the same accuracy and performance requirements at all temperatures between minus 45° F. and plus 165° F. when tested as specified in 4.5.2.

3.8.3 Intake and expulsion. The tester covered by this specification shall be capable of taking in a sufficient amount of electrolyte to obtain an accurate specific gravity reading, in a maximum of 15 seconds at 75° F. to 80° F. and shall be capable of expelling the electrolyte, in accordance with 3.5.2, in a maximum of 15 seconds at 75° F. to 80° F. when tested as specified in 4.5.6.

3.8.3.1 (for military agencies). Additionally, the class 1, style B tester shall be capable of taking in a sufficient amount electrolyte, to obtain an accurate specific gravity reading, in a maximum of 45 seconds at minus 20° F. to minus 25° F. It shall be capable of expelling the electrolyte, in accordance with 3.5.2, in a maximum of 45 seconds at minus 20° F. to minus 25° F. when tested as specified in 4.5.6.

3.9 Vibration test for case and tester class 1, style B (for military agencies). The carrying case with the instrument and floats inclosed shall be capable of withstanding subjection to a vibration test as specified in 4.5.8. At the end of this test, the instrument or its components shall not be damaged, nor shall the components be permanently dislodged within the case.

3.10 Caution note for class 2, style B. When a hard rubber inlet tube is used for class 2, style B tester, a caution label shall be fastened to the inside of the case (see 5.1.1), containing the class 2, style B tester, reading as follows:

"CAUTION. Do not use as a storage for wet tester.
"Bottom rubber plug with hard rubber projecting tube is to be used on 48-volt, lead-acid, radio B batteries."

3.11 Class and style characteristics. Unless otherwise specified, tester syringes covered by this specification shall have a specific gravity range, float size, inlet tube size, and cylinder length in accordance with table III (see 6.2).

3.11.1 Spare parts. When it is required that more than one cylinder, float, or thermometer, of the same or of different styles, are necessary to complete a test set (tester), the procuring activity may specify the items, quantity, class and style desired (see 6.2). Table III may be used as a guide in the selection of the various spare parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

TABLE III. Parts selection guide

Class	Style	Specific gravity	Float size (length)	Inlet tube (length)	Cylinder (length)	Inlet tube (inside dia.)
			Inches	Inches	Inches	
1	A	1.130 to 1.310	4-1/2 to 6	2-1/4 min.	see 3.4	
	B	1.100 to 1.350	4-1/2 to 6	2-1/4 min.	see 3.4	
	C	1.160 to 1.250	3-1/2 to 4-1/2	4 min.	8-5/8*	5/16 max.
	D	1.210 to 1.300	3-1/2 to 4-1/2	4 min.	8-5/8*	5/16 max.
2	A	1.130 to 1.310	4-1/2 to 6	2-1/4 min.	see 3.4	
	B	1.100 to 1.300	3-1/2 to 4-1/2	3*, soft rubber, 4-1/2*, hard rubber (see 3.5.2 and 3.11)	7*	
	C	1.120 to 1.300	6-3/4*	7*	10-5/8*	
	D	1.060 to 1.240	6-3/4*	7*	10-5/8*	
	E	1.200 to 1.280	6-3/4*	7*	10-5/8*	
3	A	1.130 to 1.310	4-1/2 to 6	2-1/4 min.	see 3.4	
	B	1.150 to 1.400	5*	2-1/4 min.	see 3.4	
	C	1.200 to 1.400	4-1/2 to 6	2-1/4 min.	see 3.4	

*Tolerance permitted is plus or minus (±) 3/8 inch.

3.12 Identification. Each tester shall be marked with the manufacturer's name or trademark. For the military agencies, the testers covered by this specification shall be marked for identification in accordance with MIL-STD-130.

3.13 Instructions. The contractor shall furnish two copies of a set of instructions placed within the compartment provided in the storage case with each tester supplied under this specification. The instructions shall contain necessary operating instructions, illustrations, parts list, manufacturer's part numbers, assembly and disassembly instructions, and safety precautions to insure proper operation and maintenance of the testers. Additional copies of the instructions and the mailing address to which additional copies are to be sent shall be as specified by the procuring activity (see 6.2).

3.13.1 Approval of set of instructions. When specified, a preliminary copy of the set of instructions shall be submitted to the contracting officer for approval prior to publication and delivery (see 6.2).

3.14 Workmanship. The tester shall be free from defects which will affect the operation and life of the instrument. The glass in the glass components shall be uniformly distributed and free from defects such as checks, chips, scratches, bubbles, blisters, and other defects which will affect its appearance or which may impair its serviceability. The rubber in the rubber components shall be free from defects such as abrasions, blemishes, blooms, breaks, deformations, pits, and other defects which will affect its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction inspection. Unless otherwise specified, preproduction inspection shall consist of all the tests specified in 4.5 and examination in accordance with 4.3 except that sampling in accordance with MIL-STD-105 shall not apply (see 6.2). Preproduction samples shall be as specified in 3.1. Unless otherwise specified, preproduction inspection shall be performed by the contractor (see 6.2). Failure of preproduction sample to pass preproduction inspection shall be cause for rejection.

4.2 Acceptance inspection. Provided a preproduction sample is specified, acceptance inspection shall be in accordance with 4.3 and 4.4. Unless otherwise specified, acceptance inspection shall be performed by the contractor (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

4.3 Acceptance examination. Provided a preproduction sample is specified, sampling for acceptance examination shall be in accordance with MIL-STD-105. The testers selected for examination shall be examined for conformance to the acceptable quality level (AQL) specified in 4.3.1.

4.3.1 Examination. The testers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1.1 Classification of defects. Examination shall be conducted in accordance with table IV.

TABLE IV. Categories and defects*

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component part missing (see 3.3).
102	Ballast missing or not securely fixed in position in float (see 3.3.1).
103	Stem of float not coaxial with bulb (see 3.3.1).
104	Graduated scale in stem of float missing (see 3.3.1).
105	Replacement data not marked on float scale (see 3.3.1).
106	Scale not printed in black (see 3.3.2).
107	Numbers on scale not in two or three places (sides) as required (see 3.3.2).
108	Cylinder or barrel of instrument which houses float not of clear glass (see 3.4.1).
109	Ends of cylinder not rounded (see 3.4.1).
110	Eyelet in bulb (upper fittings) missing (see 3.5.1).
111	Instrument rolls when placed on a smooth surface (see 3.5.3).
112	Glass parts touch surface when instrument laid on flat surface (see 3.5.3).
113	Numbers on correction scale missing, not easily readable, or not placed correctly (see 3.6.2 and 3.6.3).
114	Caution note for class 2, style B tester missing or not fastened to inside of case (see 3.10).
115	Spare parts missing when required (see 3.11.1).
116	Identification marking missing or not as specified (see 3.12).
117	Set of instructions missing or operating instructions incomplete (see 3.13).
118	Glass not free of checks, chips, scratches, bubbles, or blisters (see 3.14).
119	Rubber not free of abrasions, blemishes, blooms, breaks, deformations, or pits (see 3.14).

*Examination shall not be restricted to the classified possible defects listed above.

4.3.1.2 Dimensional examination. The testers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Acceptance tests (to be used only when preproduction sampling is specified). Sampling for acceptance tests shall be in accordance with MIL-STD-105, using inspection level S-3. The AQL shall be 4.0 percent defective. The samples selected shall be subjected to the tests (a), (b), and (c) below.

(a) Float accuracy (see 4.5.3.4).
(b) Thermometer accuracy (see 4.5.4).
(c) Syringe test (see 4.5.6).

4.5 Test methods.

4.5.1 Storage tests.

4.5.1.1 Low temperature test. The class 1, style B testers inclosed within the case shall be placed in an ambient air temperature of minus 60° F. to minus 65° F. for a period of 72 hours. At

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

the end of this time, the tester shall be removed and allowed to stabilize at an air temperature of plus 80° ± 10° F. The tester shall then be inspected for evidence of damage and tested for accuracy and performance in accordance with 4.5.3.4, 4.5.4, and 4.5.6 (see 3.3.3, 3.8.1, and 3.6.5).

4.5.1.2 High temperature test. The class 1, style B tester inclosed within the case shall be submitted to a 24 hour cycle of temperature starting with 10 hours at plus 90° ± 5° F., followed by 5 hours of steady increase to plus 150° to plus 155° F., 4 hours at plus 150° to plus 155° F., and 5 hours of steady decrease to plus 90° ± 5° F. The tester shall then be inspected for evidence of damage and tested for accuracy and performance in accordance with 4.5.3.4, 4.5.4, and 4.5.6 (see 3.3.3, 3.8.1, and 3.6.5).

4.5.2 Operating temperature test. The testers after being stabilized at both their low and high operating (temperature) range (see 3.8.2), and while at these temperature extremes (see 3.6.4), shall be tested for accuracy in accordance with 4.5.3.4, 4.5.4, and 4.5.6.

4.5.3 Floats. Test of the floats shall consist of measurements of overall length, ratio of scale length to overall length, axial symmetry, amount of liquid required to float the tester in the syringe furnished with it and accuracy of scale divisions.

4.5.3.1 Symmetry. Axial symmetry shall be determined when the float is floating freely in solution of the highest specific gravity for which it is intended. In this condition, the stem must not deviate by more than 1/16 inch per inch of the length of the scale (see 3.3.1 and 3.4).

4.5.3.2 Float buoyancy and position. The minimum amount of liquid required to float the float, with ranges as specified in table I, shall be determined on the instrument completely assembled and held vertically. This test may be conveniently made by measuring the amount of water required to fill the instrument to the respective marks on the float when it is resting in a fixed position. The container for the water should rest on the pan of an open laboratory balance. The minimum amount of liquid withdrawn from the container measured in grams and expressed as milliliters for a particular instrument shall not exceed the maximum amounts specified in 3.3.2, table I. When the water is expelled from the instrument, an additional weighing will indicate whether liquid retained is in excess of the amount permitted by 3.5.2.

4.5.3.3 Float accuracy. The accuracy of calibration of the float shall be determined at the upper and lower extremes of its specific gravity range and at the midpoint. The float shall be tested in a solution of sulphuric acid and water adjusted to the required specific gravity by means of a tester standard whose accuracy is attested by a certificate issued by the National Bureau of Standards.

4.5.4 Thermometer accuracy. Test of thermometers supplied as part of the syringe-tester equipment shall be made at 32°, 80°, and 120° F. by comparison with standardized thermometers used simultaneously in the same water bath. Each instrument shall be brought to an equilibrium reading at the time of comparison, and in addition each thermometer supplied as an integral part of the syringe-tester, class 1, shall be tested to determine differences in temperature indicated when first filled and the equilibrium values (see 3.6.5).

4.5.5 Tests of rubber components. Rubber parts when immersed in 1.400 specific-gravity sulfuric-acid solutions shall not roughen nor shall there by any sediment in the test solution at the end of 96 hours at room temperature (see 3.5.3).

4.5.6 Syringe test. The intake capacity and expulsion efficiency of the bulb shall be tested for compliance with the requirements of 3.5.1, 3.5.2, and 3.8.3. All testers covered by this specification, except the class 1, style B tester, shall be tested for intake and expulsion of electrolyte with the tester and electrolyte stabilized at 75° to 80° F., and then filled enough to obtain an accurate specific gravity reading in a maximum of 15 seconds (see 3.8.3), and they shall expel the electrolyte in accordance with 3.5.2 and 3.8.3. The class 1, style B tester shall be subjected to the test above and also shall be tested for intake and expulsion of electrolyte with the tester and electrolyte stabilized at minus 20° to 25° F. and then filled to overflow tube level or full enough to obtain an accurate specific gravity reading in a maximum of 45 seconds (see 3.8.3.1). It shall also expel the electrolyte in accordance with 3.5.2 and 3.8.3.

4.5.7 Thermal shock test. With the tester at a stabilized temperature, draw in 35 cc. ± 5 cc. of electrolyte that has been cooled to provide a temperature differential of 140° to 145° F. and expel. Repeat this procedure three times. Upon completion of this test, there shall be no evidence of cracking or shattering of any of the elements of the tester (see 3.4).

4.5.8 Vibration test, class 1, style B. The class 1, style B instrument with floats, inclosed in the carrying case, shall be subjected to a vibration test of 10 c.p.s. and total excursion of 0.200 ± 0.020 inch, for 30 minutes (see 3.9).

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:49Z
Check the source to verify that this is the currentversion before use.

4.5.9 Temperature differential. The testers should be filled, expelled, and refilled with a liquid at approximately 32°, 80°, and 120° F. until equilibrium is reached. Equilibrium temperature may be used for 4.5.4.

4.6 Defective units. Units found defective in the course of inspection shall be corrected to the satisfaction of the Governmnet or shall be replaced with nondefective units.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table V. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4 defects per hundred units.

TABLE V. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, not waterproof sealed. Bulging or distortion of container.
Containers	Not as specified.
Contents	Number of unit packages in intermedicate and shipping containers more or less than specified.

4.8 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.8.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.8.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.9 Alternate sampling (when preproduction sampling is not specified).

4.9.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
For dimensional examination	S-3	2.5

4.9.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Tests shall be conducted in accordance with table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

TABLE VI. Test procedures and inspection levels

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Tester	Resistance to low temperature	3.8.1	4.5.1.1	S-1	*
Tester	Resistance to high temperature	3.8.1	4.5.1.2	S-1	*
Tester	Operating temperatures	3.8.2	4.5.2	S-1	*
Float	Symmetry	3.3.1	4.5.3.2	S-1	*
Float	Buoyancy	3.3.1 and 3.5.2	4.5.3.3	S-1	*
Float	Accuracy	3.3.3	4.5.3.4	S-3	*
Float	Thermometer accuracy	3.6.5	4.5.4	S-3	*
Tester	Resistance to sulfuric acid	3.5.3	4.5.5	S-1	*
Syringe	Capacity and expulsion efficiency	3.5.1, 3.5.2, 3.8.3, and 3.8.3.1	4.5.6	S-3	*
Tester	Thermal shock	3.4	4.5.7	S-1	*
Tester	Vibration	3.9	4.5.8	S-1	*
Tester	Temperature differential	3.6.5	4.5.9	S-3	*

* Acceptance number shall be zero.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each complete tester with instructions and chart shall be packaged in a case specified in 5.1.1.4. Prior to packaging in the case the float shall be removed. The tester, float, and thermometer (when furnished separately), shall be packaged individually as specified in 5.1.1.1 through 5.1.1.3.

5.1.1.1 Tester. The tester shall be packaged in the case specified in 5.1.1.4. Special cushioned notched partitions or locking devices shall be provided to cradle the tester to prevent damage to the instrument while being stored or transported.

5.1.1.2 Float. Each float removed from the tester and spare float when required, shall be individually packaged in a box conforming to PPP-B-566, style II, type A. Cushioning material conforming to PPP-C-843 shall be used to prevent movement of the float within the box.

5.1.1.3 Thermometer. Each thermometer when furnished separately shall be packaged in a box conforming to PPP-B-566, style II, type A. Cushioning material conforming to PPP-C-843 shall be used to prevent movement of the thermometer within the box.

5.1.1.4 Case. The case shall be constructed of paperboard, solid fiberboard, or corrugated fiberboard material treated to resist acid and moisture. The case shall be designed to receive and protect each component of the tester during shipment and storage. The case shall be one piece die-cut construction with self-locking end flaps and shall have a hinged lid lengthwise of the case with full depth front and end flaps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. Each tester shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Thirty-six testers shall be packed in any of the containers conforming to the following specifications at the option of the contractor·

Specification	Type or class
PPP-B-585	Class 3 use
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class domestic

Each container shall have a case liner conforming to, and closed and sealed in accordance with MIL-L-10547. Case liners for fiberboard containers may be omitted provided they are waterproof sealed with tape in accordance with the appendix to the fiberboard box specification. The shipping containers shall be closed, sealed, and banded in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B. Thirty-six testers shall be packed in a fiberboard box conforming to PPP-B-636, class domestic.

5.2.3 Level C. The testers shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contractor or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contractor or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tester covered by this specification may be used to test the specific gravity of electrolyte lead acid storage batteries on the following equipment and applications·

(a) Automotive
(b) Aircraft
(c) Submarine
(d) Portable and radio lead acid storage batteries
(e) For measuring mixing acid

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents·

(a) Title, number, and date of this specification.
(b) Class and style tester required (see 1.2).
(c) Preproduction inspection when required and sample quantity if different (see 3.1).
(d) Inlet tube length and hard or soft rubber for class 2, style B tester (see 3.5.2).
(e) Class and style characteristics, if different (see 3.11).
(f) Additional spare parts when required (see 3.11.1).
(g) Address to which additional copies of instructions are to be sent (see 3.13).
(h) Approval of set of instructions, when required (see 3.13.1).
(i) Preproduction inspection if different (see 4.1).
(j) Responsibility for preproduction inspection and acceptance inspection, if different (see 4.1 and 4.2).
(k) Level of packaging and packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03
Defense Supply Agency - DM

Military coordinating activity:

Defense Supply Agency - DM

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.